

THE LAW OF ITERATED EXPECTATIONS

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

SPRING 2025

ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

1 The Theorem

Theorem 1.1 (The Law of Iterated Expectations). *Let $(\Omega, 2^\Omega, \mu)$ be a finite probability space. Suppose that $\Pi \subset 2^\Omega$ is a sub-algebra of the largest algebra 2^Ω . Let S be an element of Π , and let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Then*

$$\mathbf{E}[X|S] = \mathbf{E}[\mathbf{E}[X|\Pi]|S]. \quad (1)$$

2 Preliminaries

Recall that when Ω is finite, any subalgebra of 2^Ω is automatically a complete algebra. Thus Π can be generated by a finite partition $\{\Lambda^1, \Lambda^2, \dots, \Lambda^k\}$, where $\Lambda^i \cap \Lambda^j = \emptyset$ when $i \neq j$, and $\bigcup_{n=1}^k \Lambda^n = \Omega$, and $\Pi = \text{gen}(\{\Lambda^n\})$, i.e., the collection $\{\Lambda^n\}$ are the atoms of Π . Then, for any random variable X , $\mathbf{E}[X|\Pi]$ is constant on each of the sets Λ^n , that is

$$\mathbf{E}[X|\Pi](\omega) = \frac{1}{\mu(\Lambda^n)} \mathbf{E}[X \mathbf{1}_{\Lambda^n}] \quad \text{if } \omega \in \Lambda^n,$$

where $\mathbf{E}[X \mathbf{1}_{\Lambda^n}] = \sum_{\omega' \in \Omega} X(\omega') \mathbf{1}_{\Lambda^n}(\omega') \mu(\omega') = \sum_{\omega' \in \Lambda^n} X(\omega') \mu(\omega')$.¹ Writing the previous expression using indicator functions leads to the following more compact expression:

$$\mathbf{E}[X|\Pi](\omega) = \sum_n \mathbf{1}_{\Lambda^n}(\omega) \frac{1}{\mu(\Lambda^n)} \mathbf{E}[X \mathbf{1}_{\Lambda^n}].$$

3 Proof of Theorem 1.1

The right hand side of (1) is

$$\mathbf{E}[X|S] = \frac{1}{\mu(S)} \sum_{\omega \in S} X(\omega) \mu(\omega).$$

The left hand side of (1) is

$$\mathbf{E}[\mathbf{E}[X|\Pi]|S] = \frac{1}{\mu(S)} \sum_{\omega \in S} \mathbf{E}[X|\Pi](\omega) \mu(\omega).$$

Thus, it is sufficient to show that

$$\sum_{\omega \in S} X(\omega) \mu(\omega) = \sum_{\omega \in S} \mathbf{E}[X|\Pi](\omega) \mu(\omega).$$

Let $\mathcal{K} = \{n \in \{1, \dots, n\} : S \cap \Lambda^n \neq \emptyset\}$, or in words, the set of indexes for which S has non-empty intersection with the partition element Λ^n . Since $S \in \Pi$ and Π is generated by the family $\{\Lambda^n\}$, then S can

¹ $\mathbf{1}_A$ denotes the indicator function of the set A .

be written as a disjoint union $S = \bigcup_{n \in \mathcal{K}} \Lambda^n$. Then, in the previous display, if we partition the sum over $\omega \in S$ over its different components, we get the following equivalent expression:

$$\sum_{n \in \mathcal{K}} \sum_{\omega \in \Lambda^n} X(\omega) \mu(\omega) = \sum_{n \in \mathcal{K}} \sum_{\omega \in \Lambda^n} \mathbf{E}[X|\Pi](\omega) \mu(\omega), \quad (2)$$

and then it is sufficient to show that equality (2) is true.

The left hand side of (2) is:

$$\sum_{n \in \mathcal{K}} \sum_{\omega \in \Lambda^n} X(\omega) \mu(\omega) = \sum_{n \in \mathcal{K}} \mathbf{E}[X \mathbf{1}_{\Lambda^n}]$$

The right hand side of (2) is:

$$\begin{aligned} \sum_{n \in \mathcal{K}} \sum_{\omega \in \Lambda^n} \mathbf{E}[X|\Pi](\omega) \mu(\omega) &= \sum_{n \in \mathcal{K}} \sum_{\omega \in \Lambda^n} \left(\frac{1}{\mu(\Lambda^n)} \mathbf{E}[X \mathbf{1}_{\Lambda^n}] \right) \mu(\omega) \\ &= \sum_{n \in \mathcal{K}} \left(\frac{1}{\mu(\Lambda^n)} \mathbf{E}[X \mathbf{1}_{\Lambda^n}] \right) \sum_{\omega \in \Lambda^n} \mu(\omega) \\ &= \sum_{n \in \mathcal{K}} \left(\frac{1}{\mu(\Lambda^n)} \mathbf{E}[X \mathbf{1}_{\Lambda^n}] \right) \mu(\Lambda^n) \\ &= \sum_{n \in \mathcal{K}} \mathbf{E}[X \mathbf{1}_{\Lambda^n}], \end{aligned}$$

where the first equality is by the definition of $\mathbf{E}[X|\Pi](\omega)$, the second because $\mathbf{E}[X|\Pi](\omega)$ is constant over all $\omega \in \Lambda^n$, the third one because $\sum_{\omega \in \Lambda^n} \mu(\omega) = \mu(\Lambda^n)$, and the fourth one by eliminating common terms.

Then, the left-hand side is equal to the right-hand side of (2). $\square$